P1.7

$$\frac{d\langle p\rangle}{dt} = \frac{d}{dt}(-i\hbar)\int \psi^* \frac{\partial\psi}{\partial x}\,dx = (-i\hbar)\int\left\{\frac{\partial\psi^*}{\partial t}\frac{\partial\psi}{\partial x} + \psi^*\frac{\partial^2\psi}{\partial t\,\partial x}\right\}dx$$

Use S.E.

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi \Rightarrow -i\hbar\frac{\partial\psi^*}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi^*}{\partial x^2} + V\psi^*$$

$$\Downarrow$$

$$i\hbar\frac{\partial^2\psi}{\partial x\,\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^3\psi}{\partial x^3} + \frac{\partial}{\partial x}(V\psi)$$

Thus
$$\frac{d\langle p\rangle}{dt} = \int\left\{\left(-\frac{\hbar^2}{2m}\frac{\partial^2\psi^*}{\partial x^2} + V\psi^*\right)\frac{\partial\psi}{\partial x} - \psi^*\left(-\frac{\hbar^2}{2m}\frac{\partial^3\psi}{\partial x^3} + \frac{\partial}{\partial x}(V\psi)\right)\right\}$$

$$= \underbrace{\frac{\hbar^2}{2m}\int\left(\psi^*\frac{\partial^3\psi}{\partial x^3} - \frac{\partial^2\psi^*}{\partial x^2}\frac{\partial\psi}{\partial x}\right)dx}_{I_1} + \underbrace{\int\psi^*\left[V\frac{\partial\psi}{\partial x} - \frac{\partial}{\partial x}(V\psi)\right]dx}_{I_2}$$

Note that
$$\int_{-\infty}^{+\infty}\psi^*\frac{\partial^3\psi}{\partial x^3}dx = \underbrace{\psi^*\frac{\partial^2\psi}{\partial x^2}\Big|_{-\infty}^{+\infty}}_{=0} - \int\frac{\partial\psi^*}{\partial x}\frac{\partial^2\psi}{\partial x^2}dx =$$

$$= (-1)\left\{\underbrace{\frac{\partial\psi^*}{\partial x}\frac{\partial\psi}{\partial x}\Big|_{-\infty}^{+\infty}}_{=0} - \int\frac{\partial^2\psi^*}{\partial x^2}\frac{\partial\psi}{\partial x}dx\right\} = \int\frac{\partial^2\psi^*}{\partial x^2}\frac{\partial\psi}{\partial x}dx$$

$\Rightarrow I_1 = 0$. Next
$$I_2 = \int\psi^*\left(V\frac{\partial\psi}{\partial x} - \frac{\partial V}{\partial x}\psi - V\frac{\partial\psi}{\partial x}\right)dx$$
$$= \int\psi^*\left(-\frac{\partial V}{\partial x}\right)\psi\,dx$$

Thus
$$\frac{d\langle p\rangle}{dt} = \int\psi^*\left(-\frac{\partial V}{\partial x}\right)\psi\,dx = \left\langle -\frac{\partial V}{\partial x}\right\rangle$$

This is Ehrenfest's Theorem.

Classically, $\frac{dp}{dt} = F = -\frac{\partial V}{\partial x}$.

P2.1 (a) $\Psi(x,t) = \psi(x)\, e^{-iEt/\hbar}$

Suppose that $E = E_0 + i\Gamma$, then $\Psi(x,t) = \psi(x)\, e^{-iE_0 t/\hbar + \Gamma t/\hbar}$

But $\int |\Psi(x,t)|^2 dx = \int |\psi(x)|^2 e^{2\Gamma t/\hbar}\, dx =$

$= e^{2\Gamma t/\hbar} \int |\psi(x)|^2 dx$

$\hookrightarrow \infty$ for $t \to \infty$ $\Rightarrow$ <u>non-normalizable</u> state

(b) For real $V(x) = V^*(x)$

$E\psi = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi$ take complex conjugate

$E\psi^* = -\frac{\hbar^2}{2m}\frac{\partial^2\psi^*}{\partial x^2} + V\psi^*$

$\Rightarrow$ hence $\psi^*(x)$ is also a solution, with the same energy E, of the time-independent S.E.

Hence any linear combination $a\psi(x) + b\psi^*(x)$ is also solution.

In particular, we can form real $R(x) = \psi(x) + \psi^*(x)$ and imaginary $I(x) = (\psi(x) - \psi^*(x))$ combinations. Note that $i\,I(x) = i(\psi - \psi^*)$ is real as well.

(c) For <u>even</u> $V(x) = V(-x)$

$$E\psi(x) = -\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x)$$

Change variable $x = -x' \Rightarrow \frac{d}{dx} = \frac{d}{dx'}\frac{dx'}{dx} = -\frac{d}{dx'}$

but $\frac{d^2}{dx^2} = (-1)^2\frac{d^2}{dx'^2} = \frac{d^2}{dx'^2}$

$\Rightarrow E\psi(-x') = -\frac{\hbar^2}{2m}\frac{d^2\psi(-x')}{dx'^2} + V(-x')\psi(-x')$ but $V(-x') = V(x')$

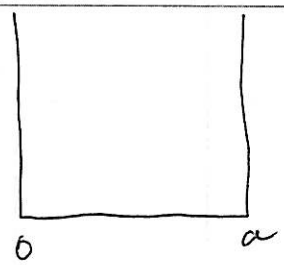
$\Rightarrow E\psi(-x) = -\frac{\hbar^2}{2m}\frac{d^2\psi(-x)}{dx^2} + V(x)\psi(-x)$.

Thus $\psi(-x)$ is also a solution, with energy E.

Hence $\psi_{even} = \psi(x) + \psi(-x)$

$\psi_{odd} = \psi(x) - \psi(-x)$

P2.3



$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi \quad \text{inside the well.}$$

$$\psi = 0 \quad \text{outside.}$$

Let us try $E=0$: $\quad -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = 0 \Rightarrow \psi(x) = A + Bx$

Boundary condition $\psi(0) = \psi(a) = 0$

$\psi(0) = B = 0$, $\psi(a) = A = 0 \Rightarrow \psi(x) = 0$ not a solution!

Next, try $E = -|E| < 0$.

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = -|E|\psi \Rightarrow \frac{d^2\psi}{dx^2} = \sqrt{\frac{2m|E|}{\hbar^2}}\,\psi = \varkappa^2\psi$$

$\varkappa^2 > 0$.

$$\Rightarrow \psi(x) = Ae^{\varkappa x} + Be^{-\varkappa x}$$

$$\psi(0) = A + B = 0 \Rightarrow B = -A$$

$$\psi(a) = Ae^{\varkappa a} + Be^{-\varkappa a} = 0 \Rightarrow A\left(e^{\varkappa a} - e^{-\varkappa a}\right) = 0$$

$$2A\sinh(\varkappa a) = 0$$

only possible for $A = 0$.

$$\Rightarrow \psi(x) = 0.$$

Thus $E=0$ and $E<0$ solutions can not be made to satisfy boundary conditions.

P2.4

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\frac{\pi n x}{a}$$

$$\langle x\rangle = \int_0^a x\,|\psi_n|^2 dx = \frac{2}{a}\int_0^a x \sin^2\frac{\pi n x}{a}\,dx = \left(x = \frac{a}{\pi n}y\right) =$$

$$= \frac{2}{a}\left(\frac{a}{\pi n}\right)^2 \int_0^{\pi n} y \sin^2 y\,dy = \frac{2}{a}\left(\frac{a}{\pi n}\right)^2 \frac{\pi^2 n^2}{4} = \frac{a}{2}$$

$$\langle x^2\rangle = \frac{2}{a}\int_0^a x^2 \sin^2\frac{\pi n x}{a}\,dx = \left(x = \frac{a}{\pi n}y\right) =$$

$$= \frac{2}{a}\left(\frac{a}{\pi n}\right)^3 \underbrace{\int_0^{\pi n} y^2 \sin^2 y\,dy}_{\frac{\pi^3 n^3}{6} - \frac{\pi n}{4}} = a^2\left(\frac{1}{3} - \frac{1}{2\pi^2 n^2}\right)$$

$\langle p\rangle = 0$ because ψ_n is stationary state (and real).

$$\langle p^2\rangle = \int \psi_n^* \left(\frac{\hbar}{i}\frac{d}{dx}\right)\psi_n\,dx = -\hbar^2 \int \psi_n^* \frac{d^2\psi_n}{dx^2}\,dx$$

but $\dfrac{d^2\psi_n}{dx^2} = -\dfrac{2m}{\hbar^2}E_n\psi_n$

$$\Rightarrow \langle p^2\rangle = -\hbar^2\left(-\frac{2m}{\hbar^2}E_n\right)\underbrace{\int_0^a \psi_n^*\psi_n\,dx}_{1} = 2mE_n = \left(\frac{\pi n\hbar}{a}\right)^2$$

$$\Rightarrow \sigma_x^2 = \langle x^2\rangle - \langle x\rangle^2 = a^2\left(\frac{1}{3} - \frac{1}{2\pi^2 n^2} - \frac{1}{4}\right) = \frac{a^2}{4}\left(\frac{1}{3} - \frac{2}{\pi^2 n^2}\right)$$

$$\sigma_p^2 = \frac{\pi^2 n^2 \hbar^2}{a^2}$$

$$\sigma_x^2\sigma_p^2 = \frac{a^2}{4}\left(\frac{1}{3} - \frac{2}{\pi^2 n^2}\right)\frac{\pi^2 n^2\hbar^2}{a^2} = \frac{\hbar^2}{4}\left(\frac{\pi^2 n^2}{3} - 2\right)$$

$$\Rightarrow \sigma_x\sigma_p = \frac{\hbar}{2}\sqrt{\frac{\pi^2 n^2}{3} - 2} > \frac{\hbar}{2} \text{ for any } n.$$

Smallest when $n = 1$.

P2.5

$$\Psi(x,0) = A\left(\psi_1(x) + \psi_2(x)\right)$$

(a) $$1 = \int |\Psi(x,0)|^2 dx = |A|^2 \int (\psi_1+\psi_2)^* (\psi_1+\psi_2)\, dx =$$

$$= |A|^2 \int \left(|\psi_1|^2 + |\psi_2|^2 + \psi_1^*\psi_2 + \psi_2^*\psi_1\right) dx = 2A^2$$

$$\Rightarrow A = \frac{1}{\sqrt{2}}$$

(b) $$E_n = \frac{\pi^2 n^2 \hbar^2}{2ma^2} = \hbar\omega \cdot n^2 \quad \text{where} \quad \omega = \frac{\pi^2\hbar}{2ma^2}$$

$$\Psi(x,t) = \frac{1}{\sqrt{2}}\left(\psi_1(x)\, e^{-iE_1 t/\hbar} + \psi_2(x)\, e^{-iE_2 t/\hbar}\right)$$

$$= \frac{1}{\sqrt{2}}\left(\psi_1(x)\, e^{-i\omega t} + \psi_2(x)\, e^{-i4\omega t}\right) =$$

$$= \frac{1}{\sqrt{a}}\left(\sin\left(\frac{\pi x}{a}\right) e^{-i\omega t} + \sin\left(\frac{2\pi x}{a}\right) e^{-i4\omega t}\right)$$

$$|\Psi(x,t)|^2 = \frac{1}{2}\left(\psi_1^* e^{i\omega t} + \psi_2^* e^{i4\omega t}\right)\left(\psi_1 e^{-i\omega t} + \psi_2 e^{-i4\omega t}\right)$$

$$= \frac{1}{2}\left(|\psi_1|^2 + |\psi_2|^2 + \psi_1^*\psi_2\, e^{-i3\omega t} + \psi_1\psi_2^*\, e^{i3\omega t}\right) =$$

$$= \frac{1}{a}\left(\sin^2\left(\frac{\pi x}{a}\right) + \sin^2\left(\frac{2\pi x}{a}\right) + 2\sin\left(\frac{\pi x}{a}\right)\sin\left(\frac{2\pi x}{a}\right)\cos 3\omega t\right)$$

(c) $$\langle x\rangle = \int_0^a x\, |\Psi(x,t)|^2 dx = \frac{1}{a}\int_0^a x\left(\sin^2\frac{\pi x}{a} + \sin^2\frac{2\pi x}{a} + \right.$$

$$\left. + 2\sin\frac{\pi x}{a}\sin\frac{2\pi x}{a}\cos 3\omega t\right) dx$$

From <u>P2.4</u> $$\int_0^a x\, |\psi_n|^2 dx = \frac{a}{2} \Rightarrow \int x \sin^2\frac{\pi x}{a} dx = \int x \sin^2\frac{2\pi x}{a} dx = \left(\frac{a}{2}\right)^2.$$

$$I = \int_0^a \frac{x}{a} \sin\frac{\pi x}{a} \sin\frac{2\pi x}{a}\, dx = a\int_0^1 x \underbrace{\sin\pi x \sin 2\pi x}_{\frac{1}{2}(\cos\pi x - \cos 3\pi x)} dx$$

$$I = \frac{a}{2} \int_0^1 x\left(\cos \pi x - \cos 3\pi x\right) dx$$

But $\int_0^1 x \cos tx \, dx = \frac{1}{t} \int_0^1 x \, d(\sin tx) = \frac{1}{t}\left(x \sin tx \Big|_0^1 - \right.$

$\left. - \int_0^1 \sin tx \, dx \right) = \frac{1}{t^2} \cos tx \Big|_0^1 = \frac{\cos t - 1}{t^2}$

$$\Rightarrow I = \frac{a}{2}\left(\frac{\cos \pi - 1}{\pi^2} - \frac{\cos 3\pi - 1}{(3\pi)^2} \right) = \frac{a}{2} - \frac{2}{\pi^2}\left(1 - \frac{1}{9}\right) =$$

$$= -\frac{8a}{9\pi^2}$$

$$\Rightarrow \langle x \rangle = \frac{1}{a}\left[\left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2\right] + 2\left(-\frac{8a}{9\pi^2}\right) \cos 3\omega t$$

$$= \frac{a}{2} - \frac{16a}{9\pi^2} \cos 3\omega t = \frac{a}{2}\left(1 - \frac{32}{9\pi^2} \cos 3\omega t\right)$$

angular frequency $= 3\omega = \frac{3\pi^2 \hbar}{2ma^2}$.

amplitude $= \frac{a}{2} \frac{32}{9\pi^2} < \frac{a}{2}$.

(d) $\langle p \rangle = m \frac{d\langle x \rangle}{dt} = \frac{ma}{2} \frac{d}{dt}\left(1 - \frac{32}{9\pi^2} \cos 3\omega t\right) =$

$$= \frac{ma}{2} \frac{32}{9\pi^2} 3\omega \sin(3\omega t) = \frac{16}{3\pi^2} ma\omega \sin(3\omega t) =$$

$$= \frac{8\hbar}{3a} \sin(3\omega t)$$

(e) from $\psi = \frac{1}{\sqrt{2}} \psi_1 + \frac{1}{\sqrt{2}} \psi_2$ it follows that

you get $E_1 = \frac{\pi^2 \hbar^2}{2ma^2}$ with probability $\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$

and $E_2 = \frac{4\pi^2 \hbar^2}{2ma^2}$ with prob. $\frac{1}{2}$. So $\langle H \rangle = \frac{1}{2}(E_1 + E_2)$